1. Kali Linux – Installation & Configuration

Kali Linux is one of the best security packages of an ethical hacker, containing a set of tools divided by the categories. It is an open source and its official webpage is https://www.kali.org.

Generally, Kali Linux can be installed in a machine as an Operating System, as a virtual machine which we will discuss in the following section. Installing Kali Linux is a practical option as it provides more options to work and combine the tools. You can also create a live boot CD or USB. All this can be found in the following link: https://www.kali.org/downloads/

BackTrack was the old version of Kali Linux distribution. The latest release is Kali 2016.1 and it is updated very often.

To install Kali Linux —

- First, we will download the Virtual box and install it.
- Later, we will download and install Kali Linux distribution.

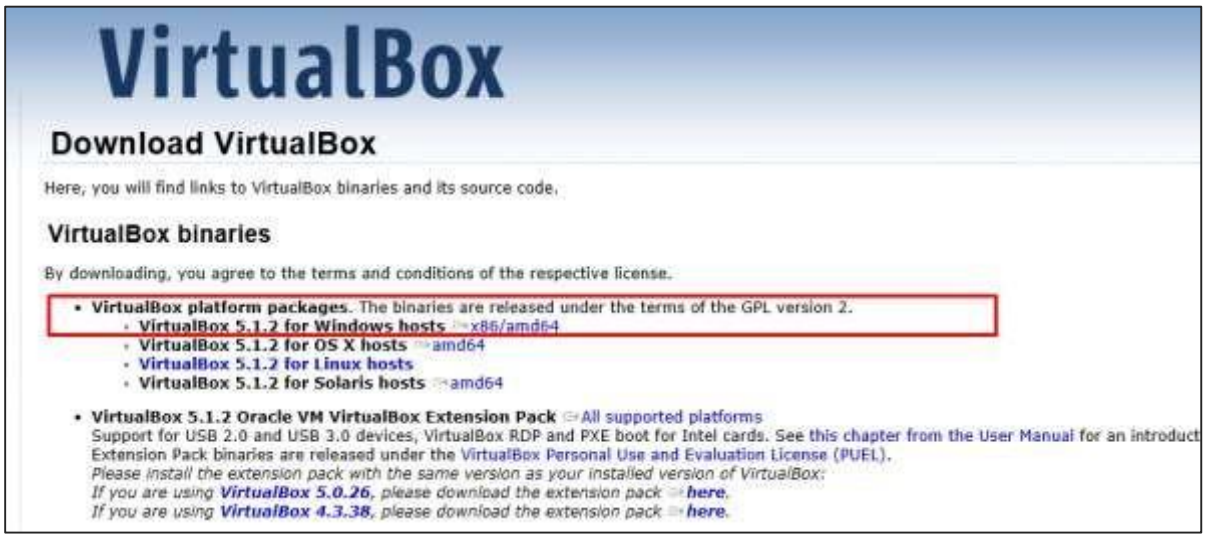
Download and Install the Virtual Box

A Virtual Box is particularly useful when you want to test something on Kali Linux that you are unsure of. Running Kali Linux on a Virtual Box is safe when you want to experiment with unknown packages or when you want to test a code.

With the help of a Virtual Box, you can install Kali Linux on your system (not directly in your hard disk) alongside your primary OS which can MAC or Windows or another flavor of Linux.

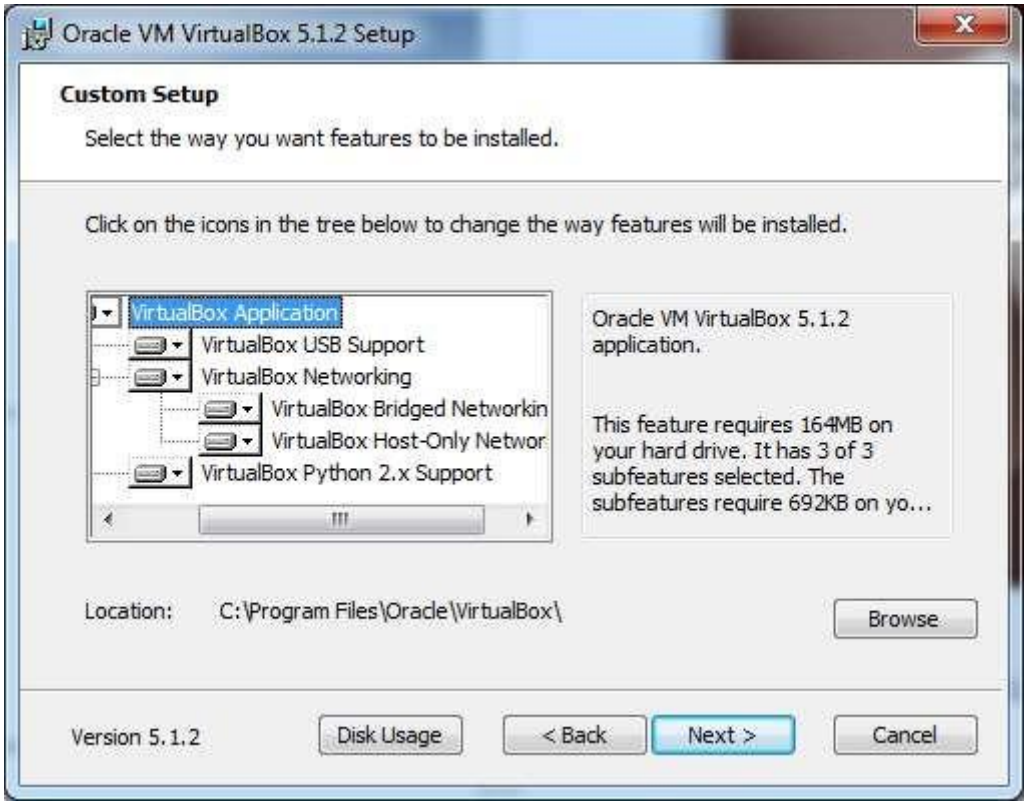
Let's understand how you can download and install the Virtual Box on your system.

Step 1: To download, go to https://www.virtualbox.org/wiki/Downloads. Depending on your operating system, select the right package. In this case, it will be the first one for Windows as shown in the following screenshot.

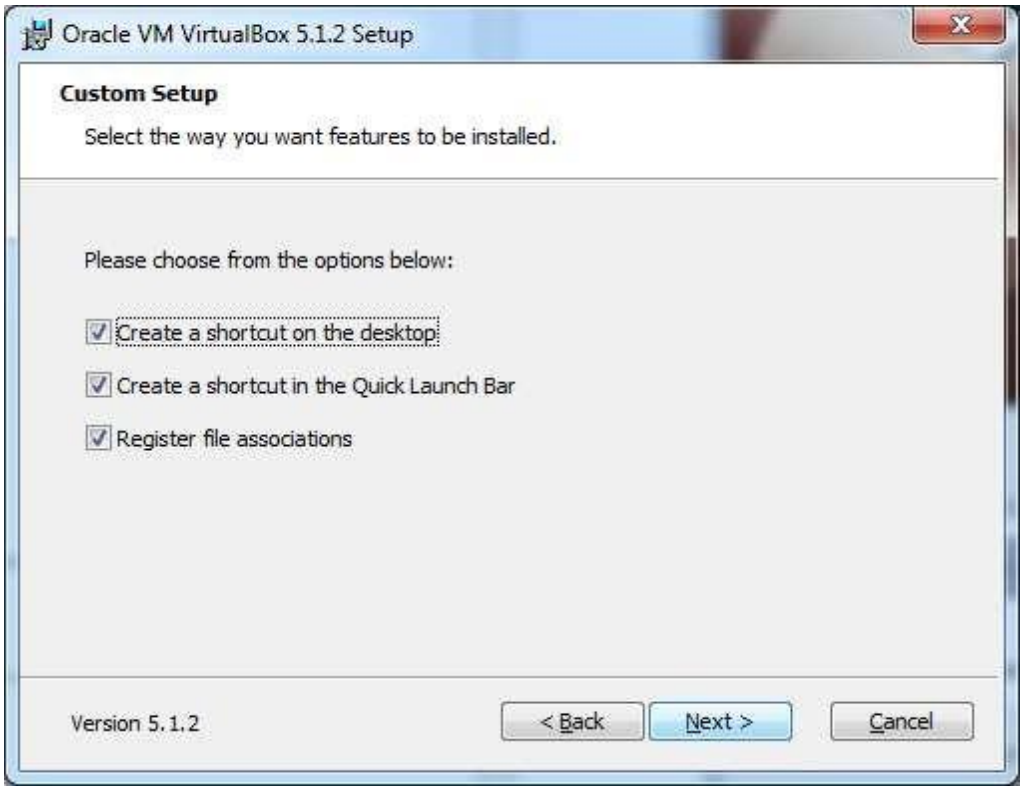


Step 2: Click **Next**.

Step 3: The next page will give you options to choose the location where you want to install the application. In this case, let us leave it as default and click **Next**.

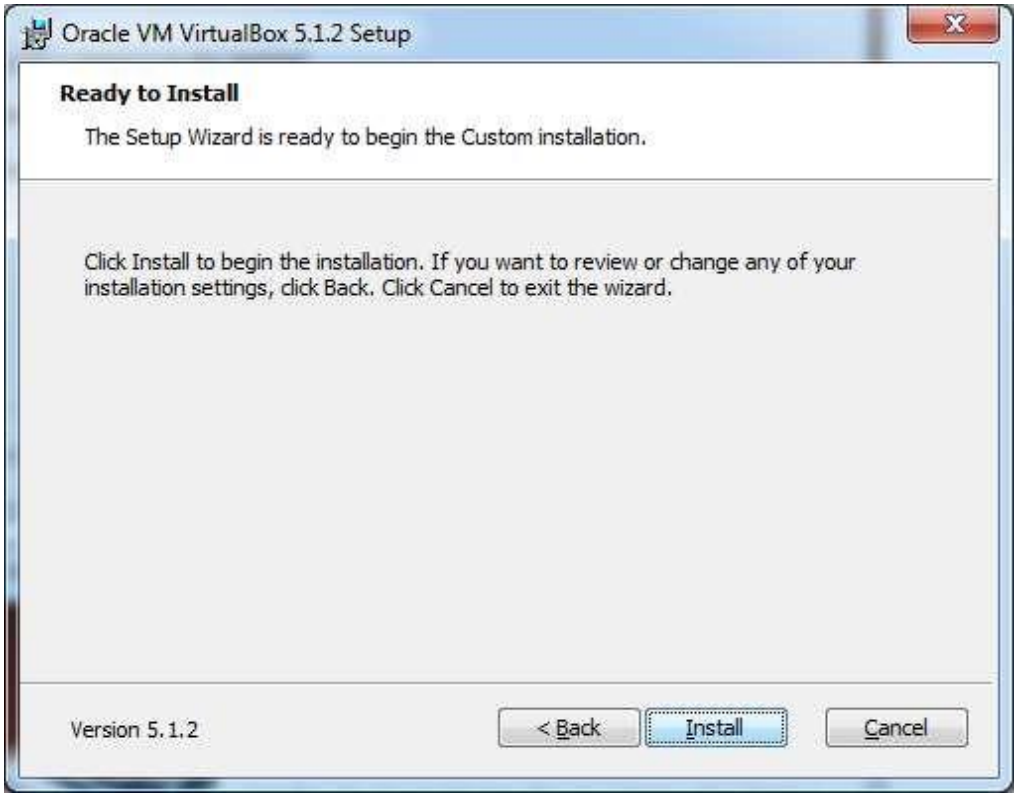


Step 4: Click **Next** and the following **Custom Setup** screenshot pops up. Select the features you want to be installed and click Next.



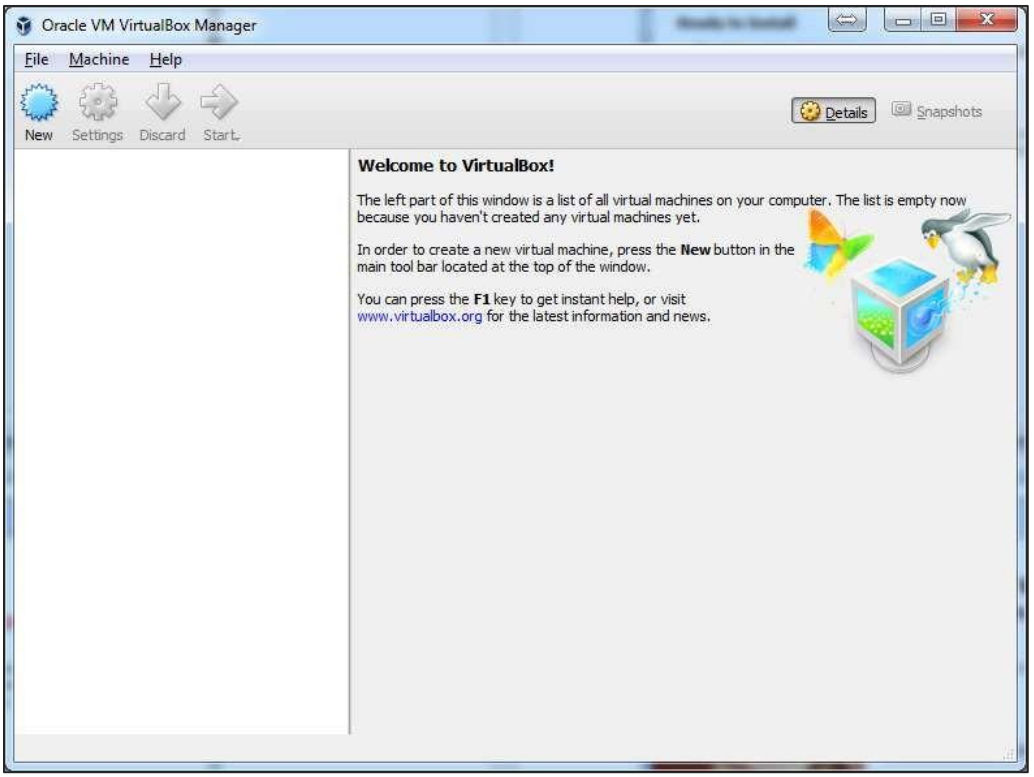
Step 5: Click **Yes** to proceed with the installation.

Step 6: The **Ready to Install** screen pops up. Click Install.



Step 7: Click the **Finish** button.

The Virtual Box application will now open as shown in the following screenshot. Now we are ready to install the rest of the hosts for this manual and this is also recommended for professional usage.



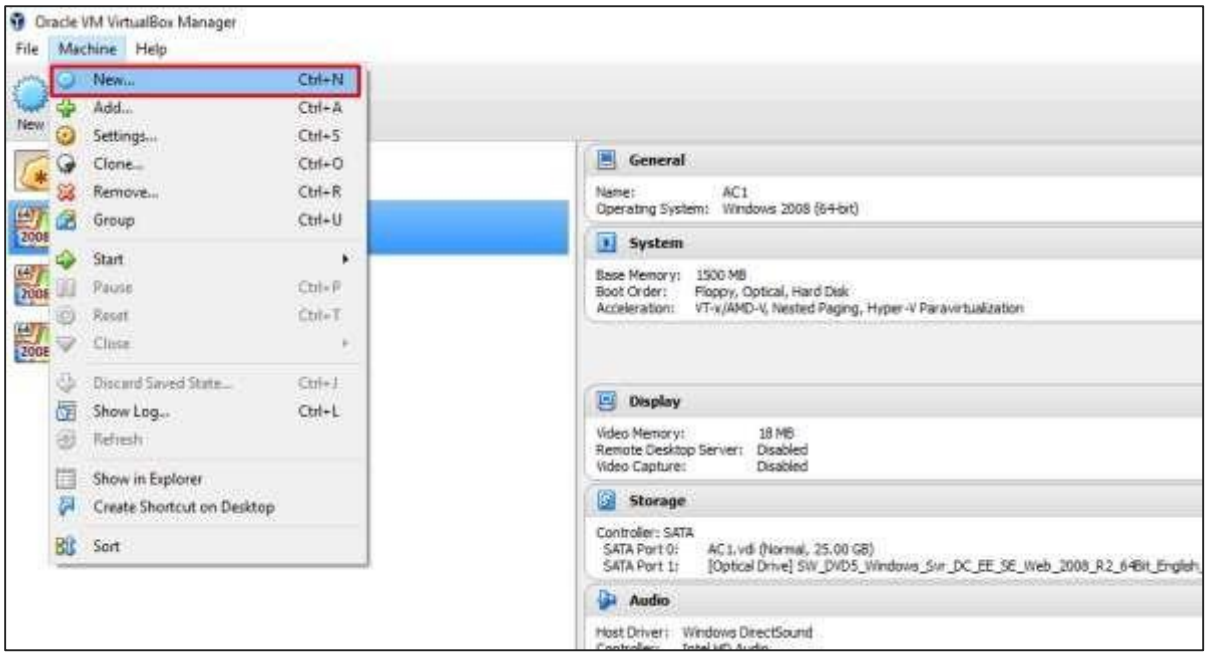
Install Kali Linux

Now that we have successfully installed the Virtual Box, let's move on to the next step and install Kali Linux.

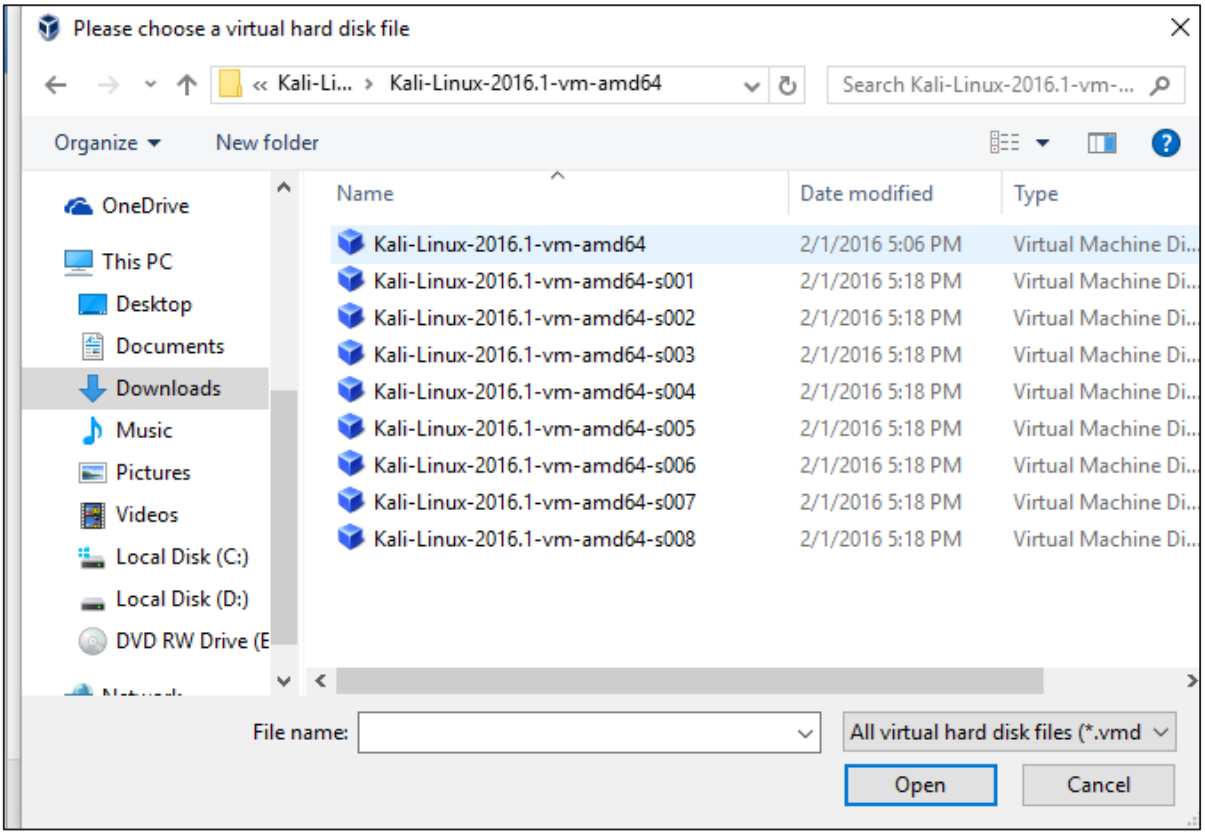
Step 1: Download the Kali Linux package from its official website: https://www.kali.org/downloads/

Image Name	Torrent	Size	Version	SHA1Sum
Kali Linux 64 bit VM	Torrent	2.0G	2016.1	2b49bf1e77c11ecb5618249ca69a46f23a6f5d2d
Kali Linux 32 bit VM PAE	Torrent	2.0G	2016.1	e71867a8bbf7ad55fa437eb7c93fd69e450f6759

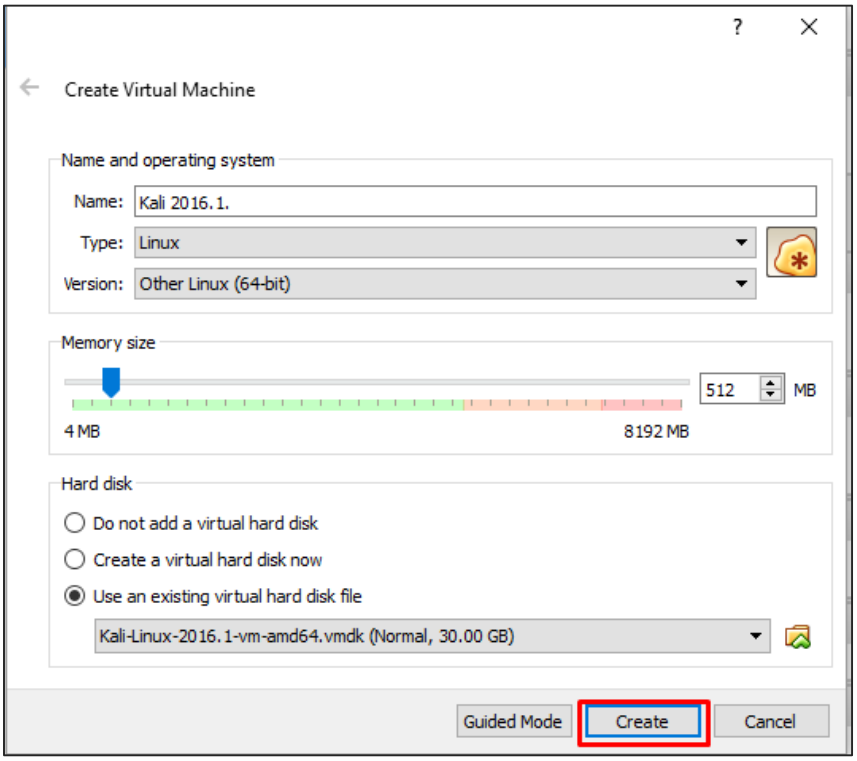
Step 2: Click **VirtualBox -> New** as shown in the following screenshot.



Step 3: Choose the right **virtual hard disk file** and click **Open**.



Step 4: The following screenshot pops up. Click the **Create** button.

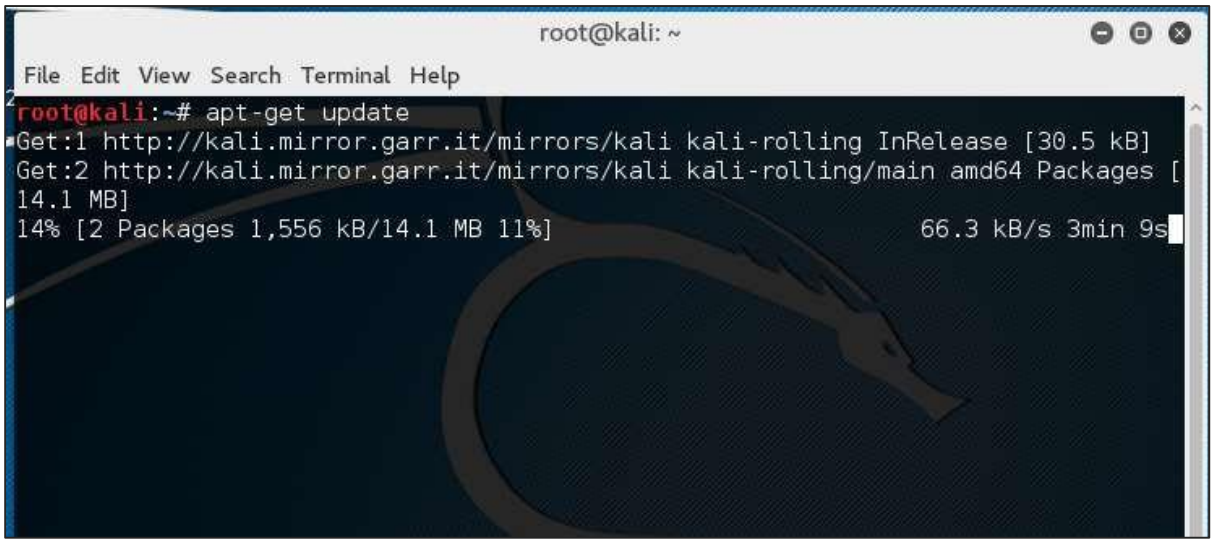


Step 5: Start Kali OS. The default username is **root** and the password is **toor**.

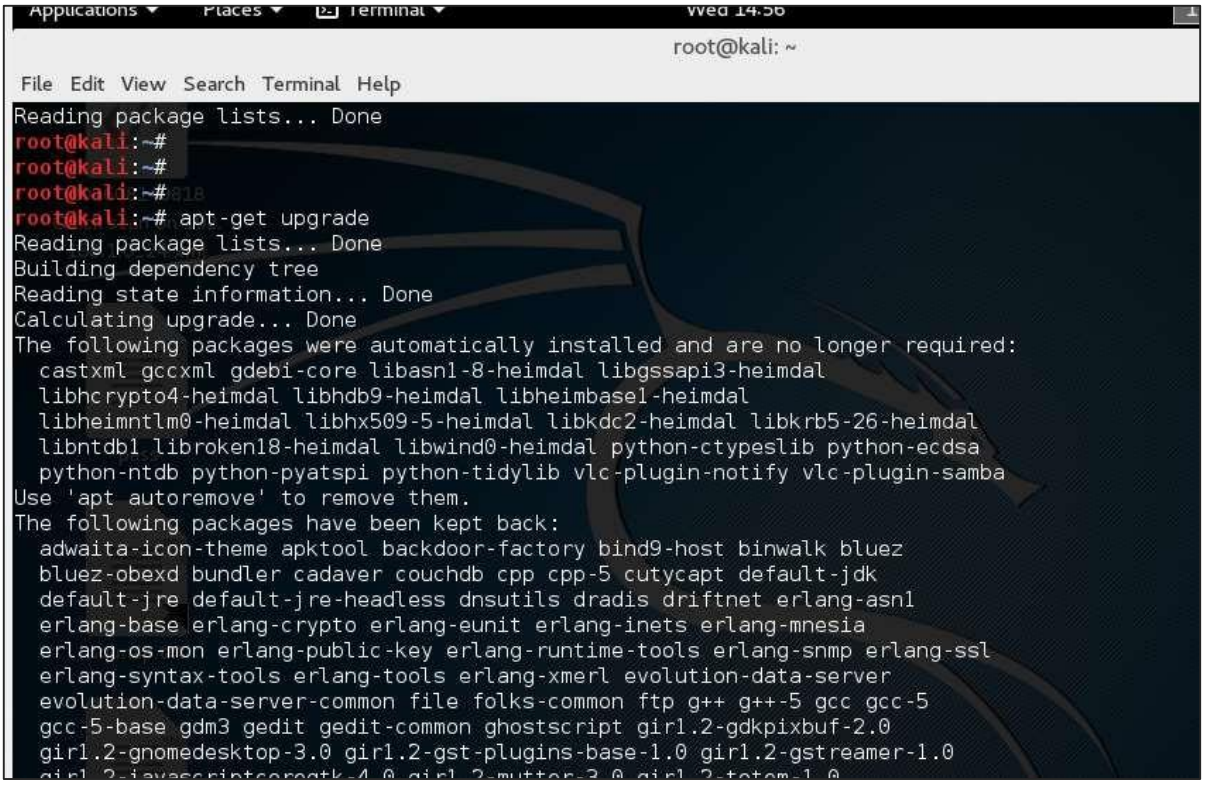
Update Kali

It is important to keep updating Kali Linux and its tools to the new versions, to remain functional. Following are the steps to update Kali.

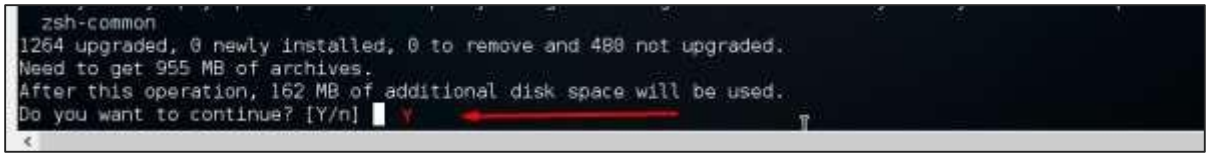
Step 1: Go to Application -> Terminal. Then, type "apt-get update" and the update will take place as shown in the following screenshot.



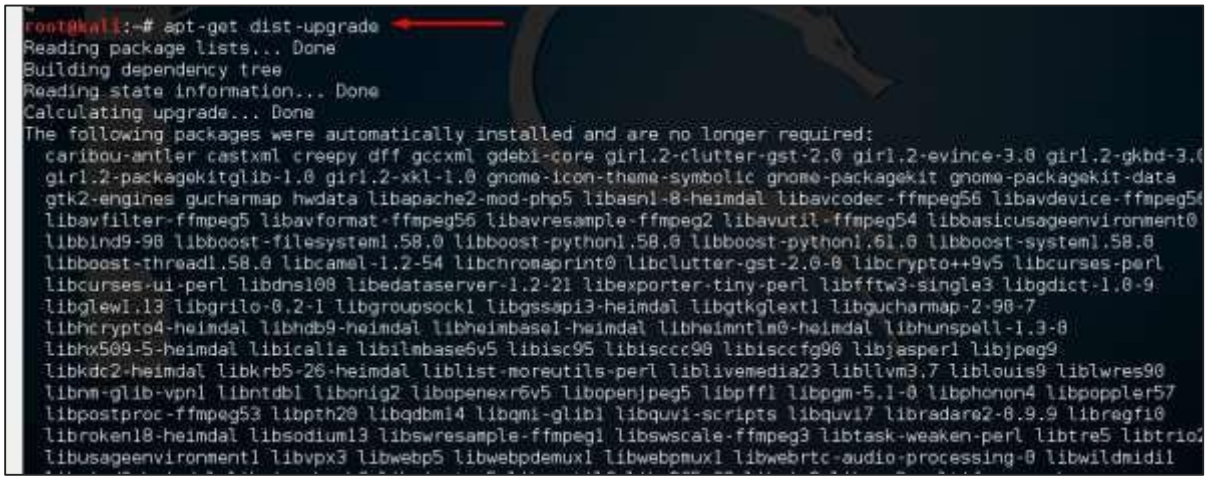
Step 2: Now to upgrade the tools, type "apt-get upgrade" and the new packages will be downloaded.



Step 3: It will ask if you want to continue. Type "**Y**" and "**Enter**".



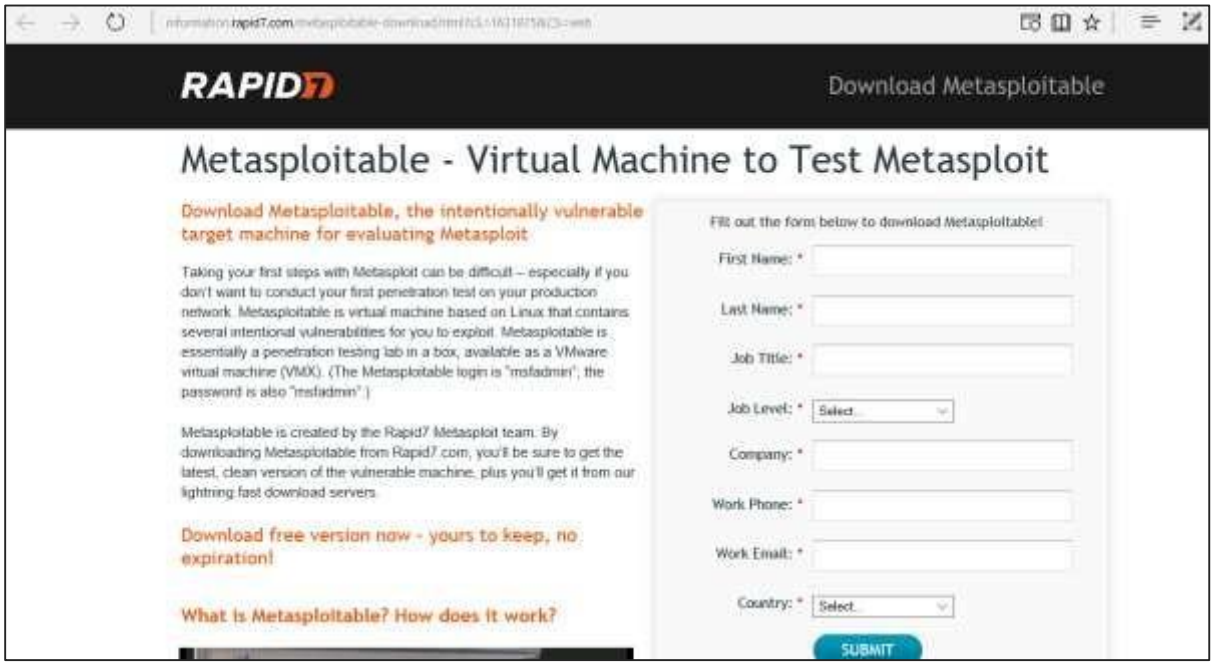
Step 4: To upgrade to a newer version of Operating System, type "**apt-get dist-upgrade**".



Laboratory Setup

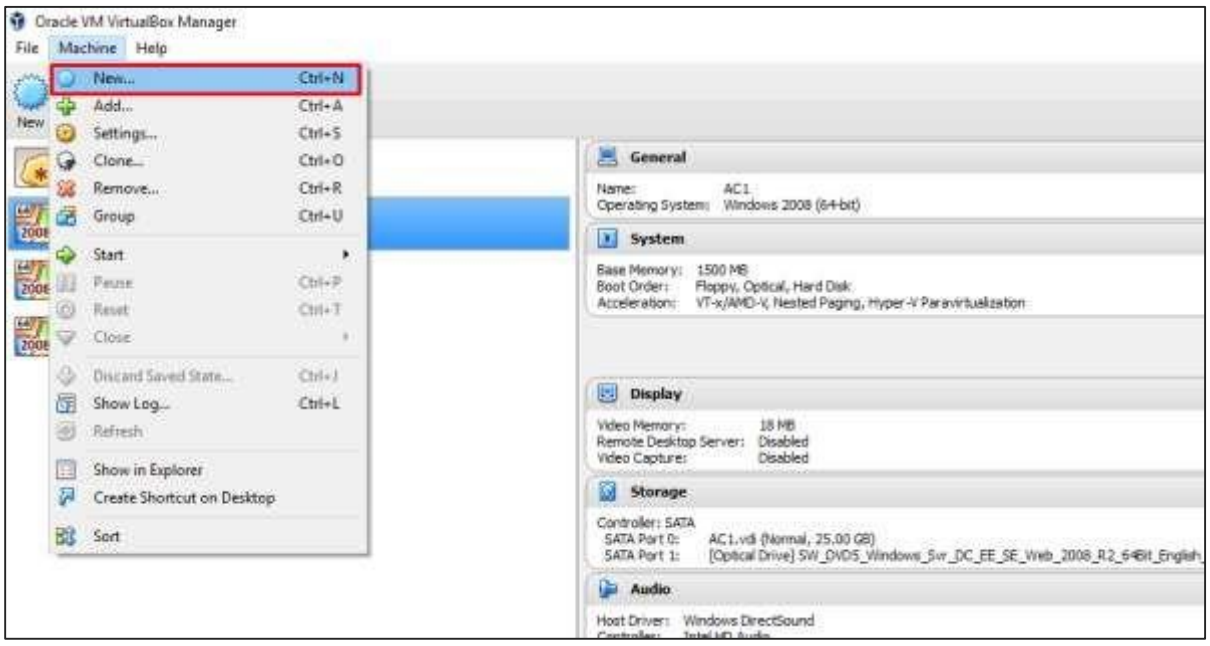
In this section, we will set up another testing machine to perform the tests with the help of tools of Kali Linux.

Step 1: Download **Metasploitable**, which is a Linux machine. It can be downloaded from the official webpage of **Rapid7**: https://information.rapid7.com/metasploitable-download.html?LS=1631875&CS=web

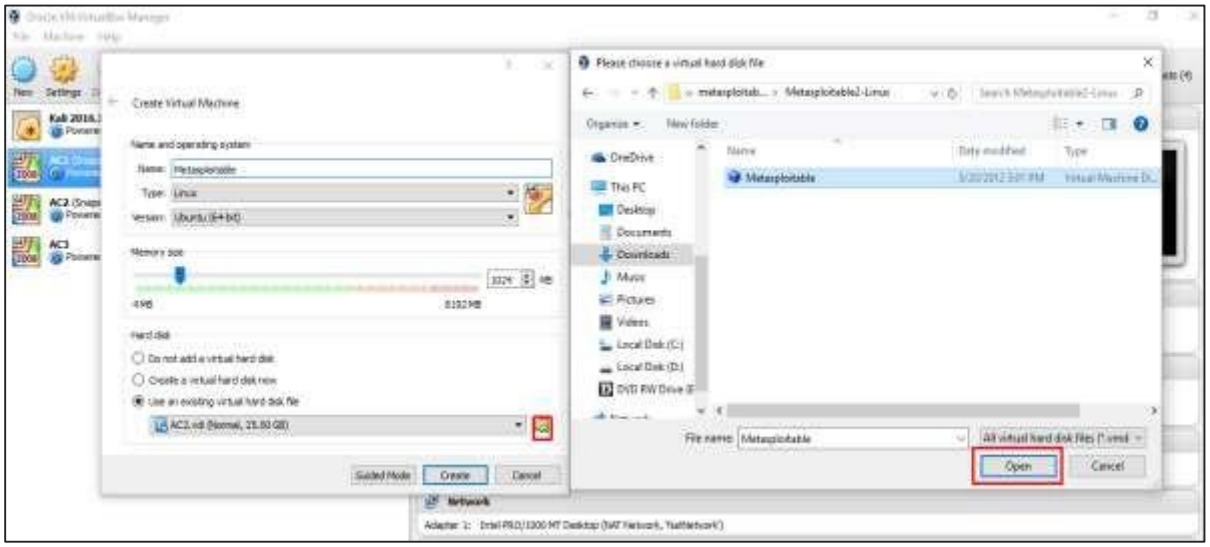


Step 2: Register by supplying your details. After filling the above form, we can download the software.

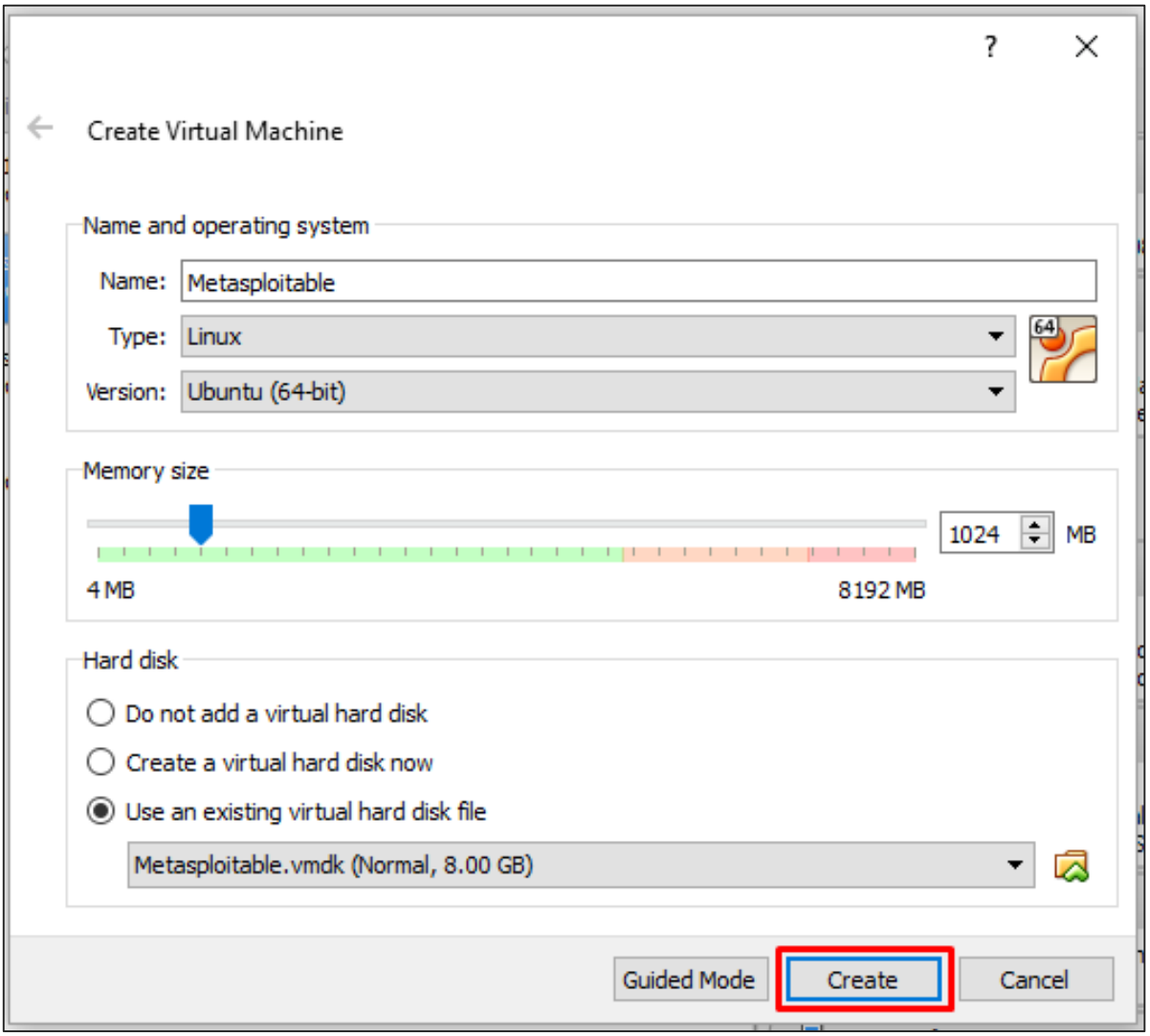
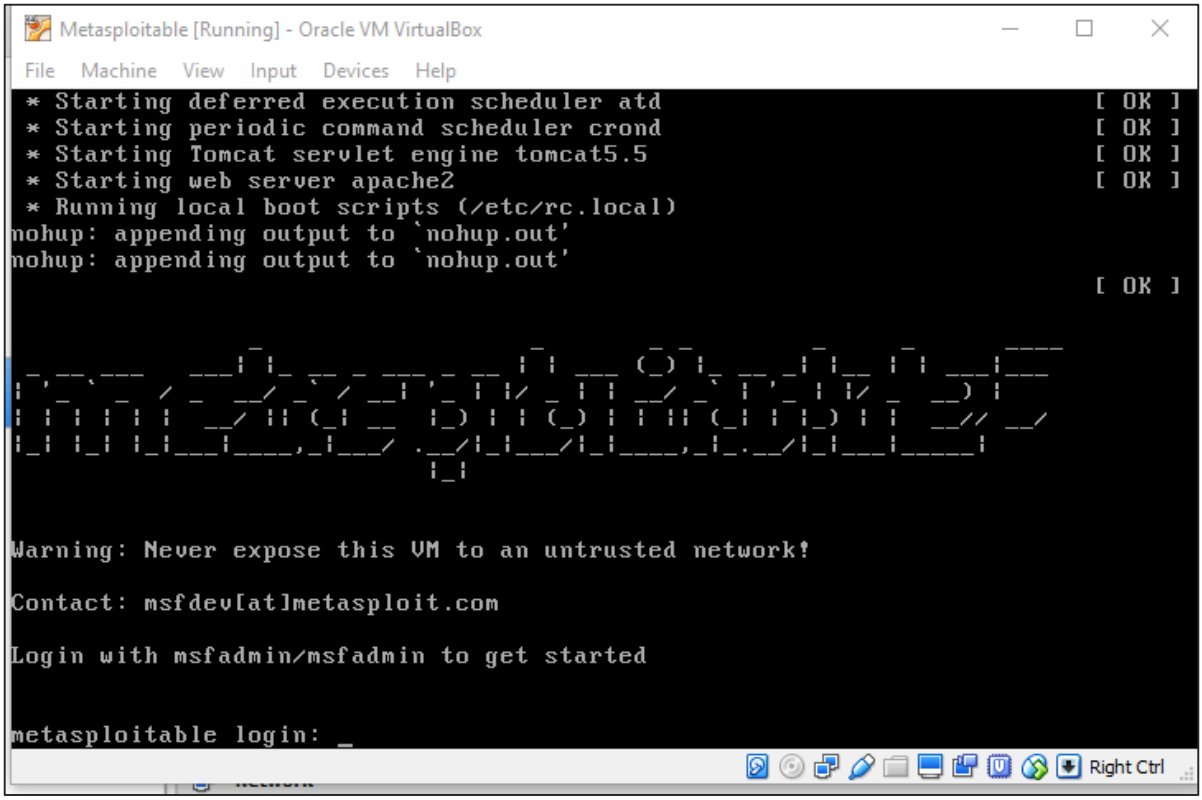
Step 3: Click **VirtualBox -> New**.



Step 4: Click "**Use an existing virtual hard disk file**". Browse the file where you have downloaded **Metasploitable** and click **Open**.



Step 5: A screen to create a virtual machine pops up. Click "Create".



The default username is **msfadmin** and the password is **msfadmin**.